

| Q.No | Scenario-Based Problem                                                                                                                                                                                                                                                                                                                              | Marks |
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| 1    | A supermarket offers discounts based on the type of customer. Create a class DiscountCalculator with overloaded methods calculateDiscount(): one accepts only the bill amount, another accepts bill amount and customer type (Regular/Premium), and another accepts bill amount, customer type, and coupon value. Display the final payable amount. | 10    |
| 2    | A courier company calculates shipping charges differently based on package details. Create a class CourierService with overloaded methods calculateCharge() to compute charges using (a) weight only, (b) weight and distance, and (c) weight, distance, and express delivery option.                                                               | 10    |
| 3    | A company stores employee details. Create a superclass Employee containing employee ID, name, and salary. Derive a subclass Manager that includes bonus and calculates the total salary. Display all employee information.                                                                                                                          | 10    |
| 4    | A hospital maintains patient records. Create a superclass Patient containing patient ID, name, and age. Derive a subclass InPatient with room number and admission days. Calculate the total hospital bill based on room charges.                                                                                                                   | 10    |
| 5    | A university manages academic records. Create a superclass Person with personal details. Derive Student from Person to store roll number and marks. Further derive GraduateStudent from Student to store project grade and determine whether the student qualifies for campus placement.                                                            | 10    |
| 6    | A vehicle manufacturing company classifies vehicles. Create a superclass Vehicle with brand and model. Derive Car to include seating capacity. Further derive ElectricCar to include battery capacity and calculate the estimated driving range.                                                                                                    | 10    |
| 7    | A banking system has a superclass Account containing account number and balance. Derive two subclasses: SavingsAccount to calculate interest and CurrentAccount to calculate service charges. Display the updated account balance for both account types.                                                                                           | 10    |
| 8    | An educational institution stores staff information. Create a superclass Staff with name and employee ID. Derive TeachingStaff to calculate teaching allowance and NonTeachingStaff to calculate overtime pay. Display the total salary for both categories.                                                                                        | 10    |
| 9    | A food delivery application estimates delivery time differently for different delivery modes. Create a superclass Delivery with a method estimateTime(). Override the method in subclasses BikeDelivery and DroneDelivery to display different delivery times based on the delivery method.                                                         | 10    |
| 10   | An online payment gateway processes different payment methods. Create a superclass Payment with a method processPayment(). Override this method in subclasses                                                                                                                                                                                       | 10    |

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|  | CreditCardPayment, UPIPayment, and NetBankingPayment to display method-specific payment processing messages and transaction charges. |  |
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